Unit 4 · Outcome 2 · Energy Sources

Sustainability principles, applied across the energy supply chain

KK07 VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.

Five stages, one resource

01 The supply chain Five stages, one resource The study design names the stages precisely: accessing → extracting → processing → transporting → using . Learn them as a sequence, because exam questions love to pin a principle to one specific stage ("evaluate the user-pays principle at the extraction stage"). The scene below follows a single energy resource down the Tarra Cove corridor in Gippsland — the fictional coastal town we have used across this chapter, sitting beside the real Latrobe Valley brown-coal fields and the Bass Strait gas province. Click

The six sustainability principles

02 The yardstick The six sustainability principles These are the same six VCAA principles you met in U3·O2 — but here you swap "environmental management" for "energy". Memorise the one-line "energy test" on each card: that is the lens you point at every stage. Two principles students confuse every year. Intergenerational equity is fairness between generations — today's energy use must not leave a degraded planet or an un-rehabilitated mine for people not yet born. Intragenerational equity is fairness within today's generation — energy access, costs and h

Principle × stage: the interactive matrix

03 The whole picture Principle × stage: the interactive matrix This grid is the dot point. Six principles down the side, five stages across the top. Click any cell to see how that principle applies at that stage, with an Australian example. This is the structure to reach for whenever a question says "apply the sustainability principles to…". Select a cell above to read how a principle plays out at a given stage of the energy supply chain.

Examples that earn marks

04 Real Australian cases Examples that earn marks Examiners reward specific, current, Australian examples over vague generalities. Lock these three — one each for the principles students find hardest to illustrate. PRECAUTIONARY Victoria permanently banned fracking and coal-seam-gas exploration and wrote the ban into the state constitution (2021) . Even though the science on groundwater risk was contested, the government acted to prevent serious, potentially irreversible harm to farmland and aquifers. That is the precautionary principle at the accessing/

Brown coal vs solar + storage

05 Compare two pathways Brown coal vs solar + storage "Compare" and "evaluate" questions want you to weigh one source against another across the principles . Toggle the two pathways and watch the pressure each one puts on each principle. Taller bar = more strain on that principle (worse for sustainability). Brown coal chain Solar + storage chain Tarra Cove gets its power from one of these. Which upholds more principles? 3D view unavailable — the principle bars below update the same way.

Which principle is being upheld?

✎ P·E·L WRITING

06 Active recall Which principle is being upheld? Drag each real-world energy action into the principle it best demonstrates. These are the kinds of stimulus snippets VCAA drops into a question — train yourself to name the principle on sight. s

P · POINT answer the verb → **E · EVIDENCE** example / mechanism → **L · LINK** tie back to question

The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Command-word decoder

◈ COMMAND WORDS

07 Read the question Command-word decoder The verb tells you what shape your answer must take. Mis-reading "evaluate" as "describe" is the single most common way to lose marks on this dot point.

Six exam traps

⚠ EXAM TRAPS

08 Don't fall for these Six exam traps Each of these has cost real students real marks. Open one, then close it and try to recall the fix.

Build a body paragraph that scores

P · E · L Build a body paragraph that scores High-band answers on this dot point follow P oint → E vidence → L ink. Make a point (name the principle & stage), give evidence (a specific Australian example with a figure), then link back to "upholding" or "undermining" the principle. Type below — the meter lights up as you hit each move. Live structure check P Point — names principle + stage 0 E Evidence — specific example / figure 0 L Link — uphold / undermine verdict 0 Keywords detected

Same question, two answers

✗ WEAK ANSWER

weak answer describes, while the

✓ STRONG ANSWER & MARKING

strong answer applies and links. ✗ Weak response — 1/3 Coal mining is bad for the environment because it digs up the land and makes pollution. We should use the precautionary principle and the user-pays principle because polluters should pay. Sustainability is important for the future so we should look after the environment for the next generation. Why

P · POINT answer the verb → **E · EVIDENCE** example / mechanism → **L · LINK** tie back to question

The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Five VCAA-style questions · 20 marks

★ PRACTICE & SELF-MARK

★ Practice exam Five VCAA-style questions · 20 marks Attempt each in your logbook first, then reveal the model answer and marking scheme. Award yourself marks honestly — the dock at the bottom tallies your total and saves it on this device. 0% Your total Mark each question above, then check your structure against the schemes. 0 / 20 marks

KK07 cheat sheet

✦ RECAP / CHEAT SHEET

≣ One-screen summary KK07 cheat sheet Sustainability principles across the energy supply chain The 5 stages (in order) Accessing — finding it & getting legal/physical entry Extracting — mining, drilling, or harvesting the flux Processing — refining, drying, enriching, converting Transporting — pipelines, transmission, ships, trucks Using — final conversion to heat / motion / electricity The 6 principles (energy test) Ecological integrity — protect ecosystems & their function Resource-use efficiency — most useful energy per unit input Intergenerational

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.

P · POINT answer the verb → **E · EVIDENCE** example / mechanism → **L · LINK** tie back to question

The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

Coal mining is bad for the environment because it digs up the land and makes pollution. We should use the precautionary principle and the user-pays principle because polluters should pay. Sustainability is important for the future so we should look after the environment for the next generation. Why it loses marks: names principles but never applies them to the extraction stage ; no Australian example; "bad for the environment" is not ecological integrity; lists three principles vaguely instead of developing one. Mark: identifies a principle (1), but no application and no evidence.

✓ STRONG / MODEL ANSWER

The user-pays principle requires those who extract a resource to bear its full environmental cost. At the extraction stage of Victorian brown coal, the operator of a Latrobe Valley mine must lodge a rehabilitation bond with the State so that the cost of returning the 135 m-deep void to a safe, stable landform is paid by the miner, not the public. This upholds user-pays by internalising the rehabilitation cost into the price of the coal — though it is only partly upheld where the bond ($289M) is below the estimated full cost ($743M). Why it scores: defines the principle (1), applies it to the named extraction stage with a specific Australian mechanism + figure (1), and links explicitly to "up

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.